Contact-Selected Virtual Mechanics with an Eight-Electrode EIT Interface

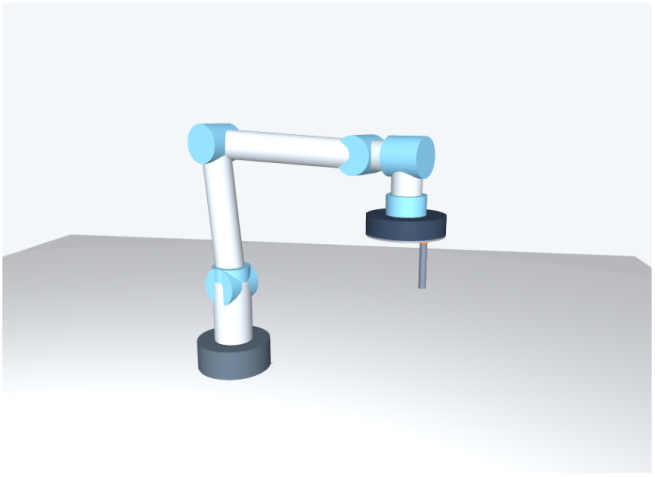
A reproducible UR3e simulation study of supported reorientation and sensing requirements

Research report prepared for Yi Zhang | 20 October 2026 | Simulation evidence only

Abstract

We examine whether a robot can use an encountered contact to organize virtual mechanics without requiring tactile force feedback. A six-joint UR3e model carries a 130 mm diameter, 33.5 mm thick device with a 90 mm active disk and eight equally spaced electrodes. A finite-element complete-electrode model generates synthetic voltages; a learned estimator selects a latched virtual attachment. Across 150 primary supported-reorientation trials, the EIT-selected and known-location controllers each complete 50/50 trials, compared with 40/50 for a fixed-centre attachment. All retain contact: the principal failure is insufficient rotation. The synthetic estimator achieves 0.91 mm median position error, while force prediction has 0.94 N mean absolute error and worsens to 1.45 N under material variation. The study contains 481 primary robot trials and 138 exploratory trials, alongside electrical calibration and reduced-model checks. These results support a controlled proof of concept for contact-location selection; the electrical transduction law and mechanical contact parameters have not been identified from the physical sensor.

(a) UR3e and rounded support: simulated geometry



(b) Eight electrodes at 45-degree intervals

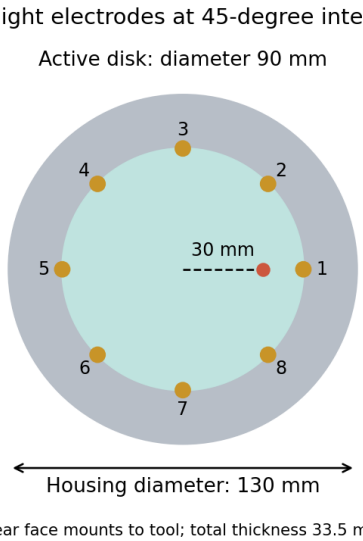


Figure 1. Implemented setup and specified sensor geometry. The arm uses official UR3e link transforms and inertia tensors, with simplified visual geometry. The support is spring-mounted in S1/S4 and rigidly fixed in S2/S3. The grey outer annulus is collidable but does not generate EIT touch observations.

1. Research question and contribution

The testable claim is that task performance can tolerate imperfect contact localization when the controller uses the contact it encounters. This is an extension of function D in the original draft [1], which already proposed spatial attachment and a latched pivot. The new evidence is a full-arm task comparison and a measurement-to-control simulation, rather than a new fifth sensing function or a claim that EIT uniquely enables contact-aware control.

2. Methodology: robot, fixture and virtual mechanics

The robot model is generated from the official Universal Robots description [2], pinned to commit d4042c1f5123c86645cc5dbd650b4f461f697ee9. Six revolute joints use the source transforms, masses, centres of mass and rotated inertia tensors. MuJoCo integrates at 1 ms with implicitfast [3]. Joint torques are limited to 54, 54, 28, 9, 9 and 9 N m. Ideal access to joint torque and exact model bias compensation are simulation assumptions; compatibility with the physical UR3e control interface remains to be verified.

The sensor rear face is fixed to tool0. Its total thickness is 33.5 mm. A 30.5 mm backing and 3 mm front collision layer are modelling choices, not measured layer thicknesses. Device mass is assumed to be 0.35 kg with a uniform-cylinder inertia approximation. A sphere of radius 8 mm forms the support. In S1/S4 it has one vertical degree of freedom, a 0.05 kg moving mass and 8 N s/m damping; S2/S3 use a fixed support. Native unilateral contact and friction allow loss of contact and sliding. No weld or pivot joint binds the tool to the support.

```
p_c = p_E + R_E r_hat; J_c = J_v - [R_E r_hat]_x J_w

f_v = K (a - p_c) - D J_c q_dot

tau = J_c^T f_v + J_w^T (K_R e_R - D_R omega) + bias - 0.2 q_dot
```

All spatial conditions share gains, torque bounds, motion references and electrical contact-event logic. Only the attachment differs: fixed at the disk centre; estimated from EIT; or supplied by the true contact geometry at the same acquired frame. A three-frame electrical dwell initiates latching. The anchor starts at the selected point and acquires a nominal 3 N preload through a 0.4 s reference ramp. This is a force demand computed from stiffness, not measured-force regulation.

Quantity	Value
Translation stiffness / damping	[1500,1500,800] N/m / [70,70,60] N s/m
Rotation stiffness / damping	S2/S3/S4: 8 N m/rad and 1.5 N m s/rad; S1: 2 and 0.4
Approach	10 mm/s; maximum 14 mm reference travel
Contact model	Friction 0.6; solref 8 ms / damping ratio 1; solimp 0.95, 0.99, 1 mm
Sensor acquisition / material lag	20 Hz nominal / first-order time constant 80 ms

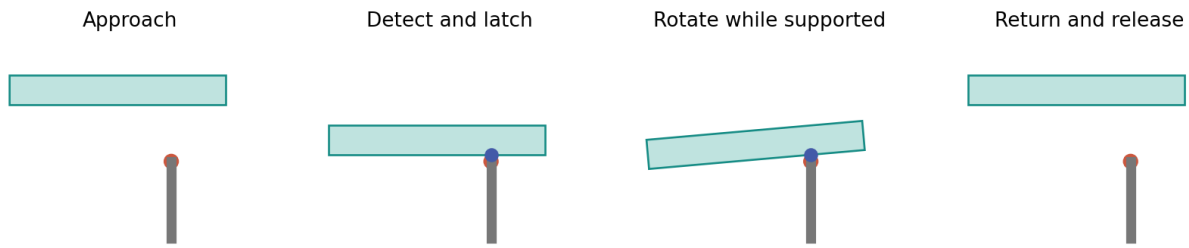
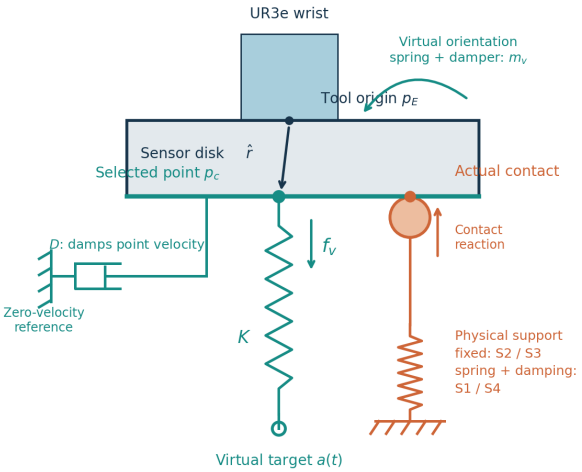


Figure 2. Task sequence. After contact acquisition, settle 0.8 s, rotate over 1 s, hold 1 s, return over 1 s, then release after 4.5 s from latch. This drawing explains the sequence; it is not a result plot.

2.1 The VMC model used in the experiments

The controller behaves as if a virtual spring pulls a selected point on the sensor toward a target in space. A virtual damper resists that point's velocity. A rotational spring and damper also turn the tool toward its requested orientation. The resulting virtual force and moment are converted into joint torques, which move the simulated UR3e.

(a) Mechanical meaning of the virtual elements



Schematic side view; teal elements exist in the controller.
The selected point can differ from the actual contact.

(b) Force, moment and joint torques

Translation: virtual spring and viscous damping

$$p_c = p_E + R_E\hat{r}, \quad v_c = J_c\dot{q}$$
$$f_v = K(a - p_c) - Dv_c$$

Orientation: rotational spring and damping

$$e_R = \mathrm{Log}(R_d R_E^T)^\vee$$
$$m_v = K_R e_R - D_R\omega$$

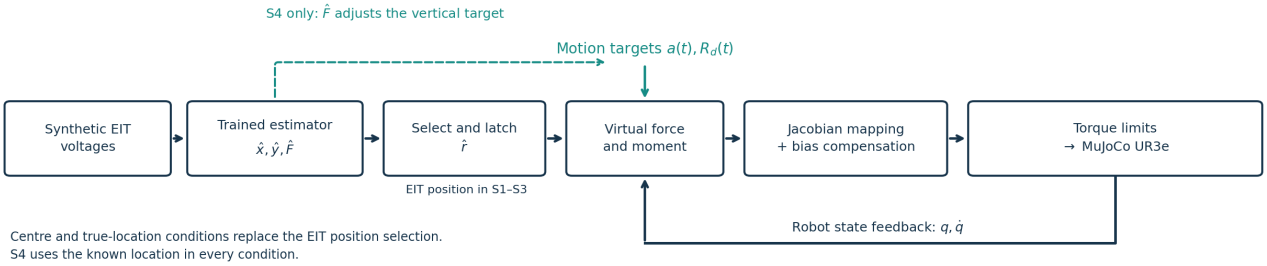
R_d: requested orientation; $\omega = J_\omega\dot{q}$

Map the virtual force and moment to six joint torques

$$\tau^* = J_c^T f_v + J_\omega^T m_v + b(q,\dot{q}) - 0.2\dot{q}$$
$$\tau = \mathrm{clip}(\tau^*, -\tau_{\max}, \tau_{\max})$$
$$\tau_{\max} = [54, 54, 28, 9, 9, 9]\ \mathrm{N\,m}$$

$b(q,\dot{q})$ compensates model gravity and velocity terms.
Rotational gains (N m/rad; N m s/rad): S1 = 2 / 0.4; S2–S4 = 8 / 1.5.

(c) Signals used by the implemented controller



Centre and true-location conditions replace the EIT position selection.
S4 uses the known location in every condition.

VMC model used in S1-S4. (a) A side view distinguishes the selected attachment from the actual support contact; their separation is exaggerated for clarity. One translational spring and damper are drawn to represent the three world-axis components. The physical support has its own mechanics. (b) Equations match the implemented controller. (c) The estimator, contact latch, VMC and torque commands form the control path. Robot state also determines the Jacobians and model bias.

Attachment and spring target. The stored offset r_hat locates the attachment on the tool; p_c is that material point's current world position. The target a(t) is a separate reference in world coordinates. At latching, it starts at the attachment's world position and then moves according to the preload and experiment schedule. The attachment stays fixed in tool coordinates while the robot moves.

Damping and orientation. Translational damping acts on the attachment's velocity J_c q_dot, and rotational damping acts on angular velocity. The spring target's velocity does not enter the damping term. Translational stiffness is [1500,1500,800] N/m and damping is [70,70,60] N s/m. The requested orientation stays constant in S1/S4 and changes during the supported rotation in S2/S3.

Where EIT enters. In the EIT conditions of S1-S3, estimated position selects the attachment at latch. The centre and true-location comparisons supply different attachment coordinates. S4 supplies the known location to every controller and compares nominal target placement with target adjustments driven by estimated or true force. The dashed path in the diagram represents the estimated-force adjustment.

3. Methodology: an explicit eight-electrode electrical model

The active surface is represented as a circular two-dimensional conductive sheet of radius 45 mm. Eight finite electrode arcs are centred at 45-degree intervals; each covers approximately 11.25 degrees. Linear triangular finite elements solve the resistive, quasi-static limit of the complete-electrode model (CEM). Finite electrode size and electrode-interface impedance distinguish this from adding arbitrary Gaussian error to a known contact position. The CEM has established use in EIT sensing models [4].

```
div(g grad(u)) = 0 inside the sheet

u + z_l g du/dn = U_l; integral_l(g du/dn) = I_l

sum_l I_l = 0; sum_l U_l = 0; g du/dn = 0 on gaps
```

Here g is sheet conductance, z is boundary contact impedance, u is the sheet potential and U is electrode potential. The nominal conductance is 0.003 S and the nominal boundary impedance is 0.02 ohm m. Adjacent current injection of 0.1 mA is combined with adjacent voltage sensing that excludes drive electrodes. Eight drives give 40 scalar channels, with reciprocal redundancy: they are not 40 independent measurements. Opposite-drive acquisition is also implemented but is not the protocol used for the reported controller results.

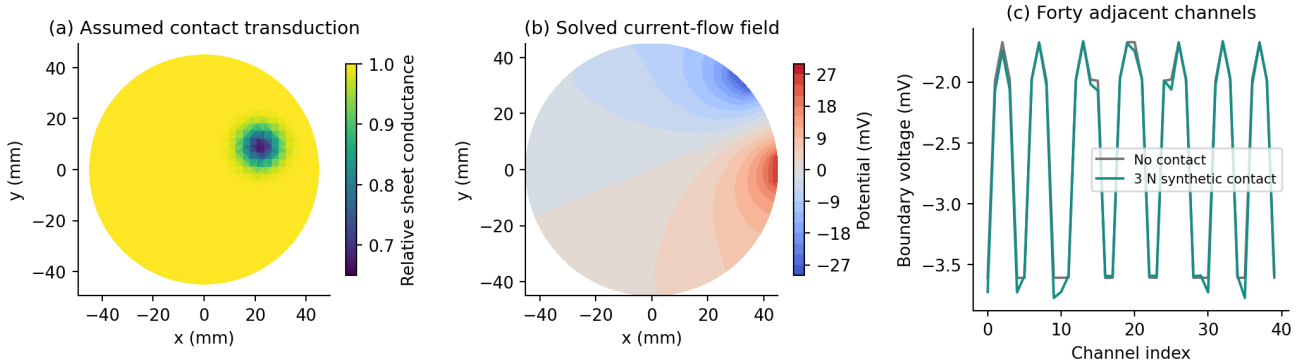


Figure 3. Electrical forward simulation for a synthetic contact at (22,9) mm and 3 N. (a) Assumed conductance perturbation. (b) The solved voltage field for one drive pair. (c) The actual electrode-voltage outputs used to construct estimator features. No inverse reconstruction is supplied with the true location.

The missing physical calibration is represented explicitly by the following phenomenological constitutive law, where a is session sensitivity and w is contact-footprint width. A negative conductance change is assumed; even its sign must be checked on the real sensor.

```
g(x,F) = g_0 [1 - 0.55 a F/(F+2) exp(-|x-r|^2/(2w^2))]
```

Contact force drives an 80 ms first-order response before voltage generation; unloading raises sensitivity by 6%. Channel noise has standard deviation 0.02% of baseline RMS voltage, shared offset drift 0.005%, and quantization is 16 bit over +/-0.2 V. These numerical settings are assumptions, not identified electronics specifications. The model is resistive rather than a frequency-dependent complex-admittance model and does not solve three-dimensional hyperelastic deformation.

4. Calibration, estimator and experiment design

A baseline is acquired for each synthetic session. Features are voltage differences divided by baseline RMS voltage. Contact is declared when their norm exceeds 0.008 for three acquired frames. Position and force are predicted by 100 extremely randomized trees; the target scales are 30 mm, 30 mm and 5 N. Leaf size is selected among 1, 3 and 6 using validation location error only. The selected size is 1. All samples of an episode, and all episodes in a session, stay within one split. The model is trained from synthetic labels; no physical EIT data are used.

Split	Episodes / frames	Sessions	Mesh spacing
Training	480 / 2400	40	4.0 mm
Validation	80 / 400	7	3.5 mm
Held-out test	100 / 500	9	2.8 mm
Material-shift test	100 / 500	9	2.8 mm

Each calibration episode has a new location within radius 37 mm and loads 1, 3, 6, 3 and 1 N, retaining loading/unloading identity. Nominal session log-standard deviations are 0.08 for conductance, 0.15 for electrode impedance and 0.12 for sensitivity; footprint width varies from 4 to 6 mm. The shift test increases these to 0.20, 0.40 and 0.35, with width 3-8 mm. Evaluation meshes also have small angular offsets. These checks reduce mesh memorization, but every split still comes from the same assumed constitutive family.

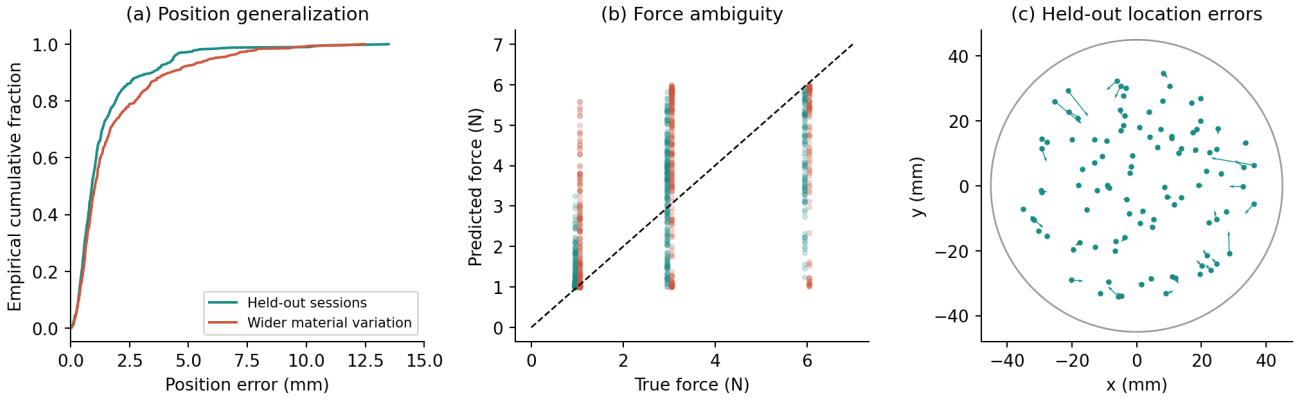


Figure 4. Held-out synthetic sensor performance. Arrows connect true contact positions to estimates for one frame per test episode. Panel (b) shows large force scatter, especially for 3 and 6 N loads; the 6 N values are often underestimated. The apparent position accuracy is not a prediction of physical TouchLoop accuracy.

Force prediction is weak even in the held-out synthetic test: 6 N contacts average 4.23 N predicted, and force R-squared is 0.48. Under wider material variation, force R-squared falls to -0.09, below a constant mean-force predictor on that split. These results do not establish reliable EIT force feedback.

Stage	Executed comparison	Paper role
S0	Original reduced model and step refinement	Original D mechanism
S1	3 offsets x 3 controllers x 5 seeds = 45	New POC 1: contact mechanics
S2	5 offsets x 3 controllers x 2 directions x 5 seeds = 150	New POC 2: main task
S3	196 localization, timing, noise and contact trials	New POC 3: sensing requirements
S4	2 spring constants x 3 targets x 3 conditions x 5 seeds = 90	Original A force-regulation branch

5. Experiments: does contact location affect unwanted motion?

Purpose. Test whether applying virtual spring and damping forces at the point where the sensor touches a support reduces unwanted tilting and displacement. If contact occurs away from the disk centre, applying these forces at the centre can create a turning moment. The virtual attachment is the selected point on the sensor where these controller forces act. This section first checks that geometric effect in a simple model (S0), then tests it with the simulated UR3e arm (S1).

S0: what is done. Represent the tool as a body that can move vertically and tilt in a plane. Apply a constant 5 N load at a point away from its centre. Compare virtual forces attached at the centre with forces attached at the known load location, and measure vertical displacement at the contact and tool rotation. This establishes the expected mechanical effect before introducing arm dynamics or EIT estimation.

The simplified model has mass m=2 kg, rotational inertia I=0.02 kg m squared, vertical stiffness K=200 N/m and damping D=30 N s/m, rotational stiffness K_phi=2 N m/rad and rotational damping 0.3 N m s/rad. Its motion equations are integrated with a fourth-order Runge-Kutta method (RK4). For a constant load, the predicted steady contact displacement is:

```
y_c = F/K + (r_hat-r)^2 F/K_phi
```

Here F is the applied force, r is its true offset from the tool centre, r_hat is the attachment offset chosen by the controller, and y_c is vertical contact displacement. The second term predicts additional displacement when the chosen attachment misses the contact location.

At the original 60 mm offset, mean displacement is 34.000 mm for fixed-centre attachment and 25.007 mm for known-location attachment; rotation is -8.594 and 0.005 degrees. The 60 mm case lies outside the new sensor radius and is retained only for draft reproduction. At an attainable 30 mm offset, the excess static displacement reduces from 9 mm to 2.25 mm. Halving the S0 step changes the contact trace by at most 0.0089 mm.

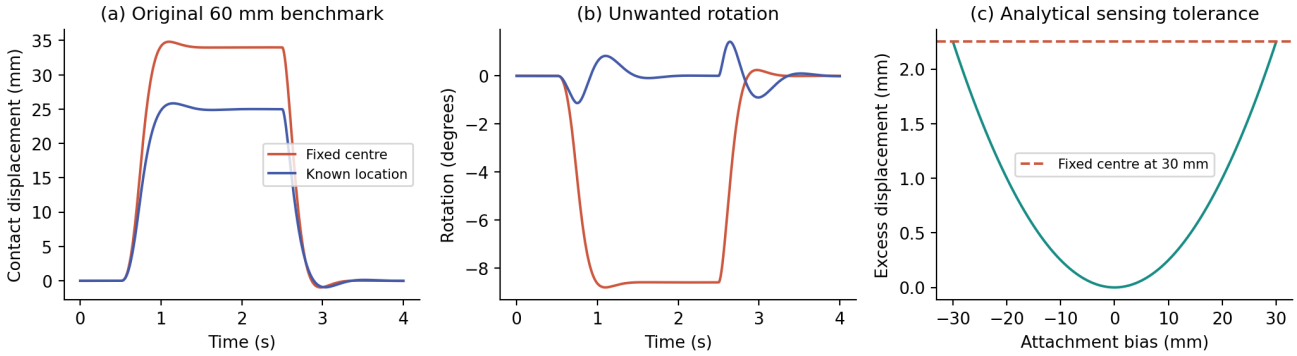
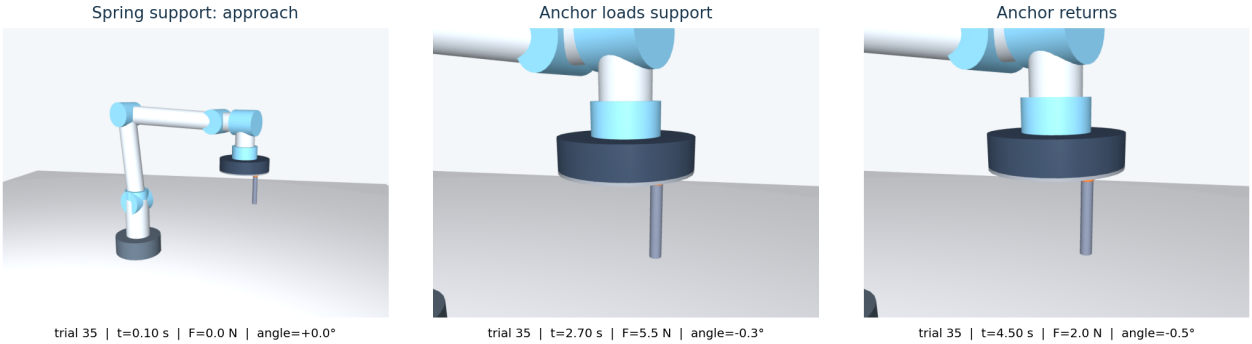


Figure 5. Reproduced reduced-model traces and the analytical bias relation. Panel (c) is a theoretical curve, explicitly separated from executed full-arm results.

S1: purpose and procedure. Test whether the same reduction in unwanted tilting appears when the full UR3e presses against a compliant support. The arm brings the sensor disk onto a rounded tip mounted on a 1000 N/m vertical spring. Contact is placed 30 mm to either side of the disk centre or at the centre. After establishing contact and a small initial load, the controller moves its virtual spring target 8 mm downward and then returns it. The robot is asked to hold its initial orientation throughout.

Compare three attachment choices: disk centre, position estimated from EIT, and true simulated contact location. Three contact positions, three controllers and five sensor seeds give 45 trials. Measure unintended rotation and the displacement between the controller target and the contacted material point per unit contact force. These measurements test the mechanical effect of attachment placement. Across off-centre cases, mean peak hold rotation is 3.71 degrees (fixed), 0.41 degrees (EIT), 0.26 degrees (known location).



MuJoCo view of S1, trial 35 (+30 mm, EIT attachment). The full arm approaches a spring-supported tip; close views show loading and return. Images reconstruct saved joint and support positions at the labelled times. S0 above is a planar reduced model and has no UR3e scene.

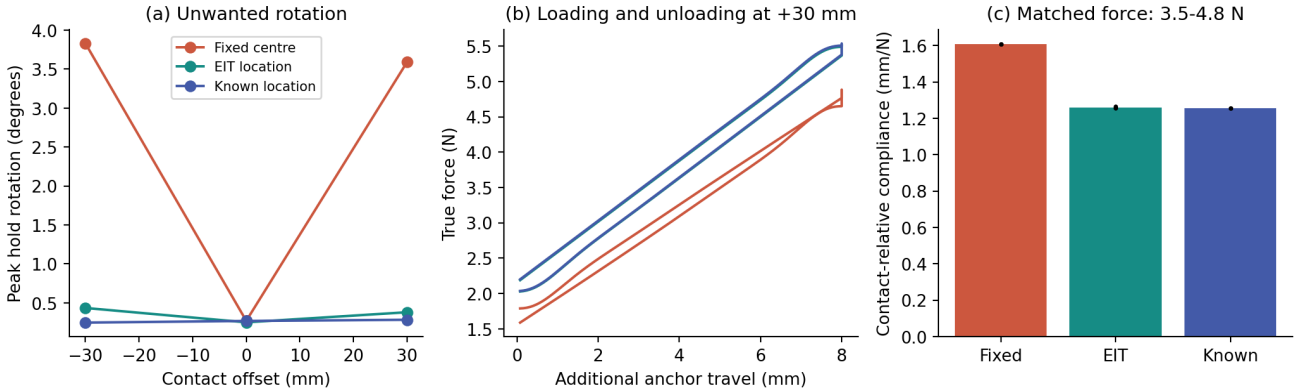


Figure 6. Full-arm mechanism experiment. Loading/unloading is shown at +30 mm. Contact-relative compliance is fitted over the same measured-force interval, 3.5-4.8 N, using separation between the moving virtual anchor and the latched material point. It is not an estimate of the object material stiffness. The material point is recovered from saved joint states; its latch coordinate is sampled at 100 Hz.

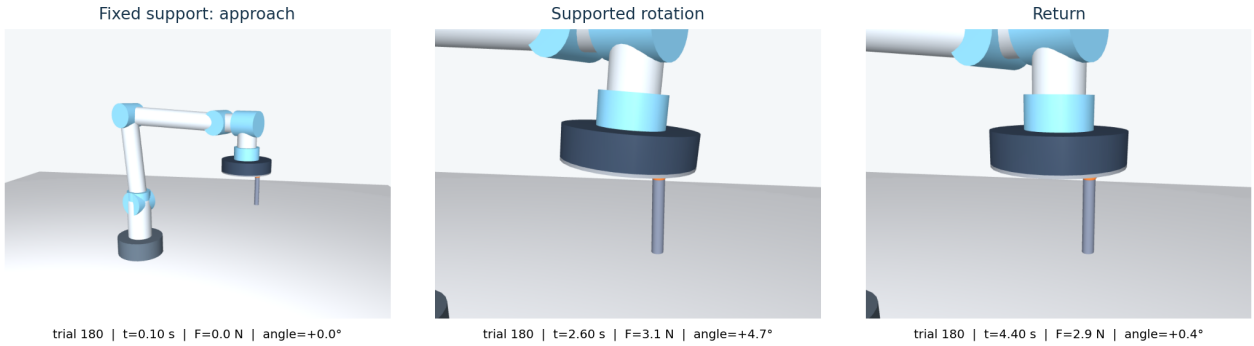
The contact-relative compliance at +30 mm is 1.608 mm/N (fixed), 1.260 mm/N (EIT), 1.255 mm/N (known location). The comparison supports the geometric mechanism before interpreting the application task.

6. Experiments: contact, latch and supported reorientation

Purpose. Test whether selecting the attachment from EIT position estimates helps the robot complete a useful motion: tilt the sensor disk to a requested angle while maintaining contact with a support. This tests whether the mechanical effect in Section 5 improves task performance. The EIT condition receives an estimated contact position; force estimates do not enter its control feedback.

What is done (S2). The UR3e approaches a fixed rounded tip until the electrical signal confirms contact. The controller then stores the selected attachment point, a step called latching. It establishes a small initial load, rotates the disk by +5 or -5 degrees, holds the angle, returns, and lifts away. Repeat at five contact offsets (-30, -15, 0, +15 and +30 mm) with centre, EIT-estimated and true-location attachments. Five sensor seeds for each direction and condition give 150 trials. Gains and motion commands are shared across the attachment choices.

How success is judged. Success requires mean commanded-axis orientation error no greater than 1 degree during 1.9-2.7 s after latch, contact force above 0.2 N for at least 99% of the 0.8-2.8 s task window, peak force no greater than 10 N and latched material-point drift no greater than 2 mm in that window. A 25 N or excessive joint-speed guard terminates a trial. Tolerances were fixed after the pilot and before the primary trial schedule.



MuJoCo view of S2, trial 180 (+30 mm, +5 degrees, EIT attachment). The fixed rounded support remains under the disk as the arm rotates and returns. Labels report simulated time, true normal contact force and achieved rotation from the saved trace.

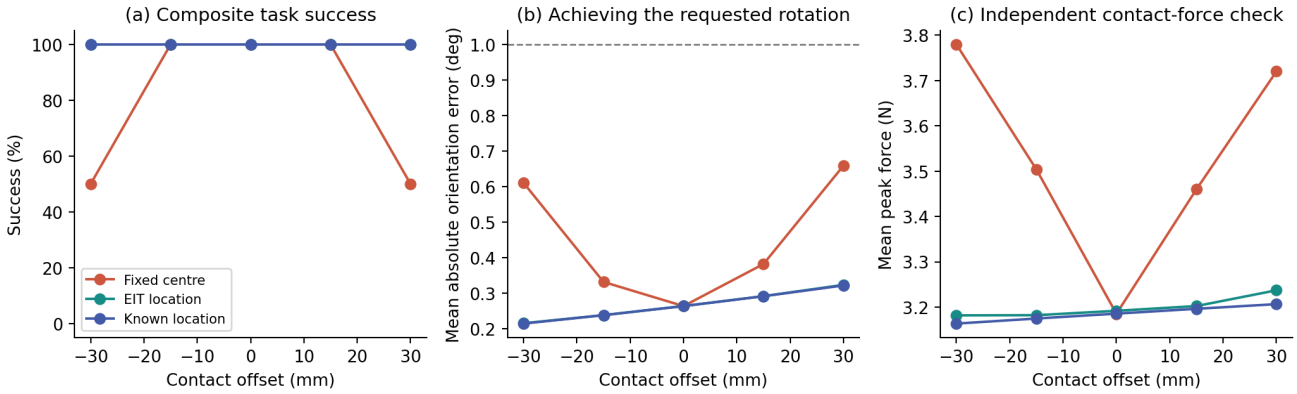


Figure 7. Main result. Each offset/controller combines two rotation directions and five sensor seeds (10 trials). At either 30 mm edge, the fixed-centre controller fails in one rotation direction. EIT and known-location success curves overlap. The same gains and task limits are used throughout.

Controller	Successful trials	Mean orientation error	Maximum anchor drift
Fixed centre	40/50	0.450 deg	0.049 mm
EIT location	50/50	0.267 deg	0.045 mm
Known location	50/50	0.267 deg	0.045 mm

An independent full three-dimensional orientation check on 474 latched reorientation and robustness trials changes no orientation pass/fail classifications; maximum off-axis rotation is 0.297 degrees. Saved

per-trial checks are included with the results.

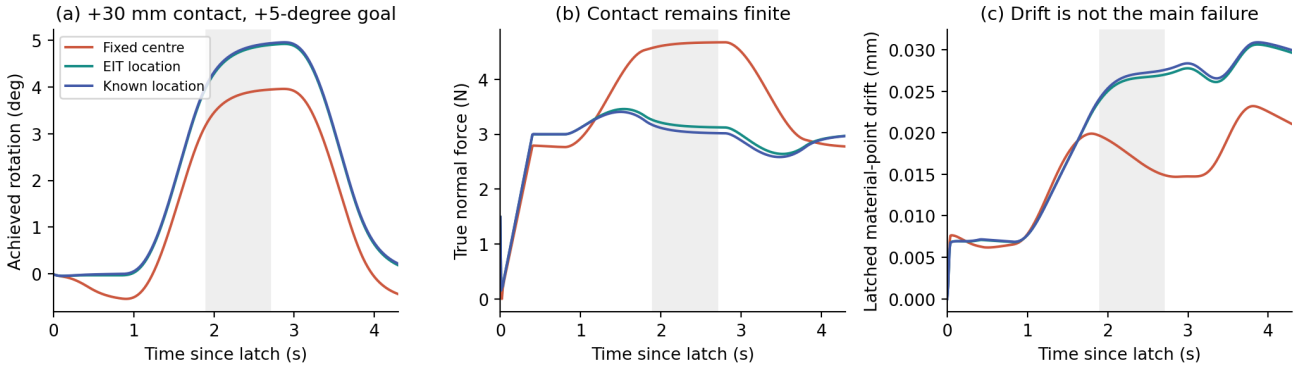


Figure 8. Representative +30 mm / +5-degree episode, seed 710. Grey shading identifies the orientation evaluation window. The fixed-centre controller retains contact and exhibits little anchor drift, yet undershoots the orientation goal. Thus contact retention or drift alone would conceal task failure.

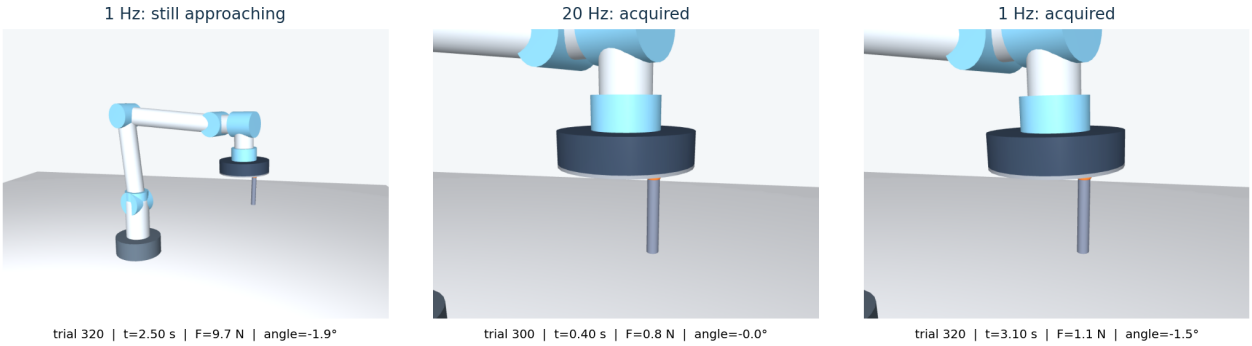
All primary S2 trials retain contact throughout the evaluated motion. Maximum contact migration in sensor coordinates is below 0.74 mm and maximum accumulated tangential slip within the task is below 0.035 mm. These are different observables: a contact locus can migrate on a rounded support even with little slip. The benefit here is the achieved motion under finite virtual mechanics, not preventing dramatic sliding.

7. Experiments: how much sensing quality is necessary?

Purpose. Determine how inaccurate, coarse or slow the contact information can become before the supported rotation task deteriorates. A second question is whether eventual task success hides excessive contact force while the robot waits for a touch reading.

What is done (S3). Repeat the approach, latch and 5-degree rotation from Section 6 with contact 27 mm from the disk centre. First supply the true contact location with a deliberately added position error, so the effect of attachment error can be measured directly. Then run separate EIT conditions that round estimated positions to coarse spatial bins, delay readings, reduce the number of readings per second, add voltage noise, or increase variation in the synthetic sensor material. Compare task success, time to latch and force during approach across 196 trials. The rotation trials follow the success criteria in Section 6.

The tested bin widths are 5, 10 and 20 mm; added delays are 0, 20, 50 and 100 ms; acquisition rates are 1, 5, 10 and 20 Hz; added normalized voltage-noise levels are 0.001, 0.003 and 0.01. A 44 mm radial clamp bounds the attachment: positive biases near the edge can be clipped, so commanded bias and realized position error are both saved. Separate stress tests combine friction coefficients 0.2/0.8 with contact time constants 4/16 ms to examine sensitivity to the assumed contact mechanics.



MuJoCo view of S3 at +27 mm and +5 degrees, EIT attachment. Trial 320 at 1 Hz is still approaching at 2.50 s with 9.7 N contact force; trial 300 at 20 Hz acquired contact at 0.40 s; trial 320 acquires at 3.10 s. These are saved states from the acquisition-rate comparison, not a new trial.

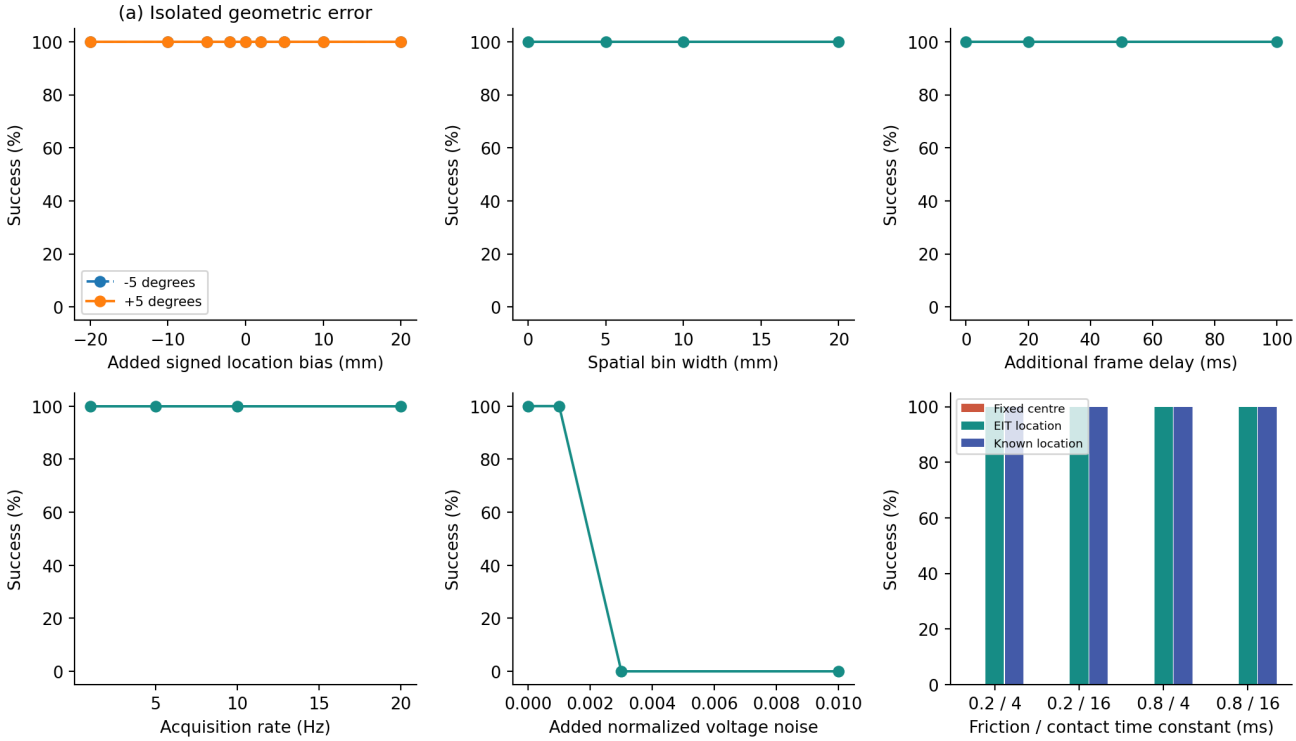


Figure 9. Primary sensing and contact robustness tests. Bias results use five paired seeds per direction; coarseness, delay, rate and noise use five seeds each. Contact-parameter groups use three seeds per controller. A successful narrow test is evidence only for its specified motion and loading regime.

All primary bias cases from -20 to +20 mm complete the small-angle task, as do the tested spatial bins. This establishes a tolerance range for this setup, not a universal minimum resolution for tactile control. Noise that causes a first electrical latch before physical contact is counted as a failed acquisition, rather than being filtered using ground-truth contact. An audit of every unique initial history and corrected affected trials is included in results/audit.

Acquisition rate	Mean time to latch	Maximum pre-latch force
20 Hz	0.30 s	2.48 N
10 Hz	0.40 s	2.48 N
5 Hz	0.60 s	3.75 N
1 Hz	3.00 s	9.73 N

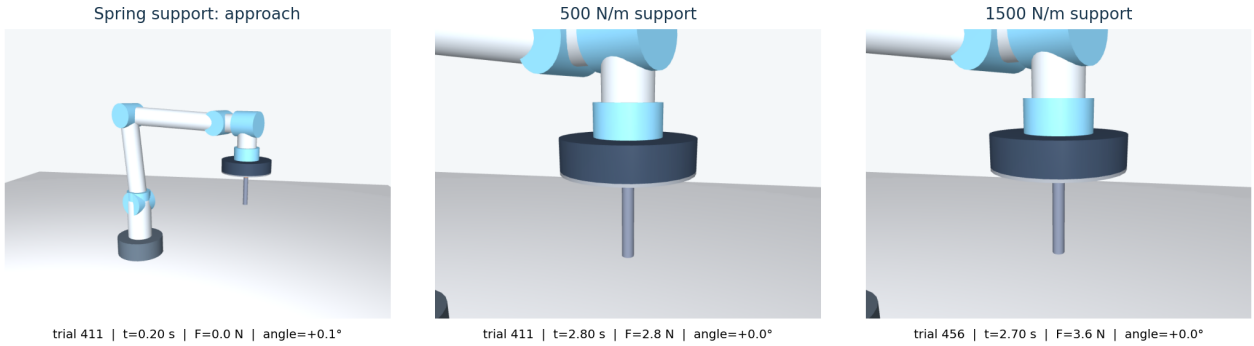
Slower sensing can still finish the eventual pivot but substantially delays acquisition and increases force during continued approach. The task-window success score alone therefore does not establish satisfactory approach behaviour. The hardware version should stop or slow the approach while confirming a tentative contact event.

8. Experiments: the optional force-regulation branch

Purpose. Test whether the learned EIT force estimate is accurate enough to help the robot maintain a requested pressing force. Figure 4(b) shows substantial force-estimation errors, so improved regulation must be checked directly. This experiment also compares EIT feedback with the performance available when exact simulated force is supplied.

What is done (S4). The UR3e presses the disk onto a rounded tip mounted on a vertical spring and holds its orientation. Request constant forces of 2, 4 and 6 N against supports with stiffnesses of 500 and 1500 N/m. Compare three conditions: nominal VMC sets a spring-target displacement from the desired force; EIT feedback adjusts that target using predicted force; true-force feedback makes the same adjustment using exact simulated force. All three receive the known contact location to isolate force estimation. Two support stiffnesses, three targets, three controllers and five sensor seeds give 90 trials. Measure actual force error and peak force.

The feedback conditions adjust the virtual spring target with an integrator gain of 0.0015 m/(N s), speed limit 3 mm/s and travel limit +/-8 mm. The true simulated contact force is used for evaluation in every condition; only the explicitly labelled true-force reference receives it as feedback. The primary error window is 1.9-2.7 s after latch.



MuJoCo view of S4 with a 4 N target and EIT force feedback. Trial 411 approaches the 500 N/m spring support, then trial 411 and trial 456 show the 500 and 1500 N/m supports during regulation. The saved force values in each panel are true simulated contact forces.

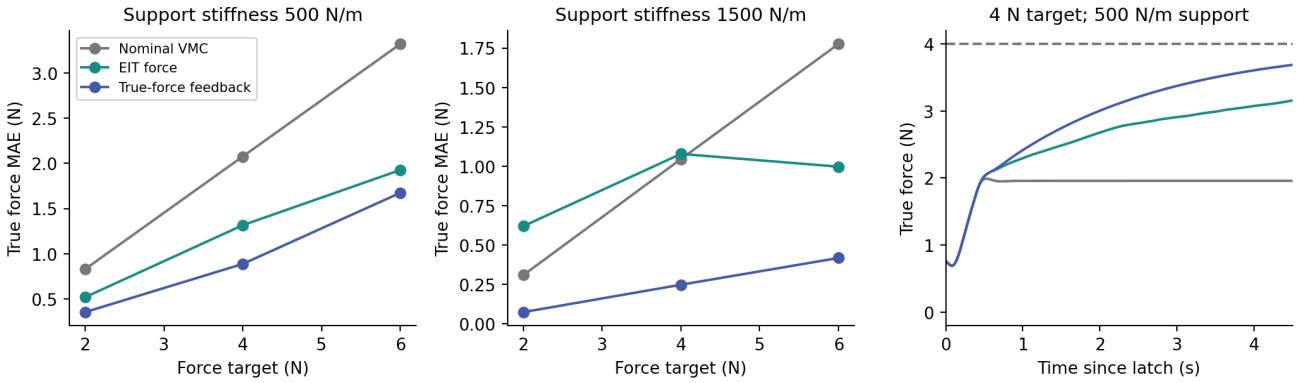


Figure 10. Force-regulation results. Each point averages five independent sensor seeds. The primary force-error window is 1.9-2.7 s after latch. The right panel shows an executed 4 N episode against the softer support. Imperfect EIT force feedback is not equivalent to a load cell.

Condition	Primary-window MAE	Late-window MAE	Whole-episode peak
Nominal VMC	1.560 N	1.560 N	4.34 N
EIT force	1.076 N	1.155 N	8.34 N
True-force feedback	0.609 N	0.269 N	5.93 N

The late window (3.8-4.4 s) is a secondary analysis added to distinguish incomplete regulation from estimator bias. True-force feedback improves as it settles, whereas EIT feedback retains appreciable

error and can produce larger forces. Averaging across targets hides individual successes and failures; the per-target plots and all trial traces are included.

In the independent electrical test, force MAE is 0.94 N and increases to 1.45 N under wider material variation. In comparison, position median error changes from 0.91 to 1.04 mm. Force and contact-footprint sensitivity are partially confounded by this model. This supports treating accurate force feedback as a capability to establish experimentally rather than a prerequisite for the new paper.

A post-hoc decoder check compares simpler calibrations on the same held-out datasets. The conditioned polynomial uses labelled training locations and estimated test locations; its ridge penalty is selected on validation. These decoders did not replace the estimator in the executed controller trials.

Force decoder	Held-out MAE	Material-shift MAE
Signal magnitude / isotonic	1.108 N	1.383 N
Location-conditioned polynomial	0.983 N	1.488 N
Joint position-force trees	0.940 N	1.451 N

What this does not test

Function B, object-stiffness identification, is not executed; it needs independently defined object deformation. Function C supplies acquisition here, but this is not a comprehensive switching-controller benchmark.

9. Exploratory follow-up and numerical verification

Purpose. Examine where the earlier benefit stops: can the robot complete larger rotations, and how large can an attachment-position error become before the task fails? These follow-up tests were added after the primary experiments, so their results are exploratory.

What is done. Repeat the supported rotation task at +10, -10, +15 and -15 degrees, with contact at -30, 0 and +30 mm. Compare centre, EIT-estimated and true-location attachments with three new sensor seeds, giving 108 trials. In another 30 trials, command larger negative attachment biases of -30, -40 and -50 mm at the 27 mm contact offset, then request +5 or -5 degrees. The attachment still obeys the active-disk clamp. Compare achieved rotation and task success using the earlier gains and pass criteria. The larger motions and errors extend the primary 5-degree task and +/-20 mm bias sweep.

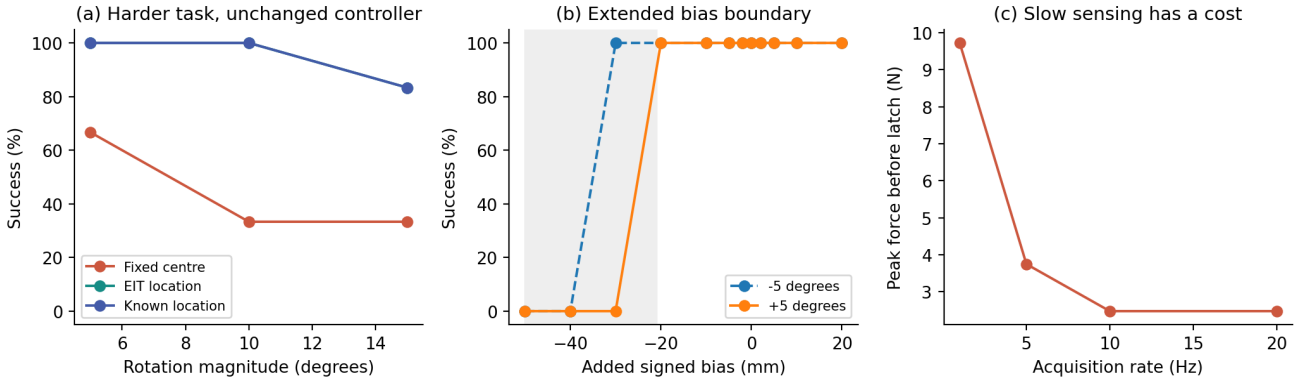


Figure 11. Follow-up experiments. (a) The 5-degree points use only the same three offsets as the harder-angle follow-up; their sample count differs. (b) Grey shading marks the newly explored negative-bias region. Positive bias is limited by the active-disk clamp. (c) Whole-approach loading reveals the cost hidden by eventual task success.

Rotation magnitude	Fixed centre	EIT location	Known location
10 degrees	6/18	18/18	18/18
15 degrees	6/18	15/18	15/18

The known-location controller also fails some 15-degree cases, identifying a task/controller limitation rather than a sensing-only failure. A -30 mm attachment bias passes in one direction and fails in the other; -40 and -50 mm biases fail in both directions. The tolerance depends on motion direction and available torque balance, not solely Euclidean localization error.

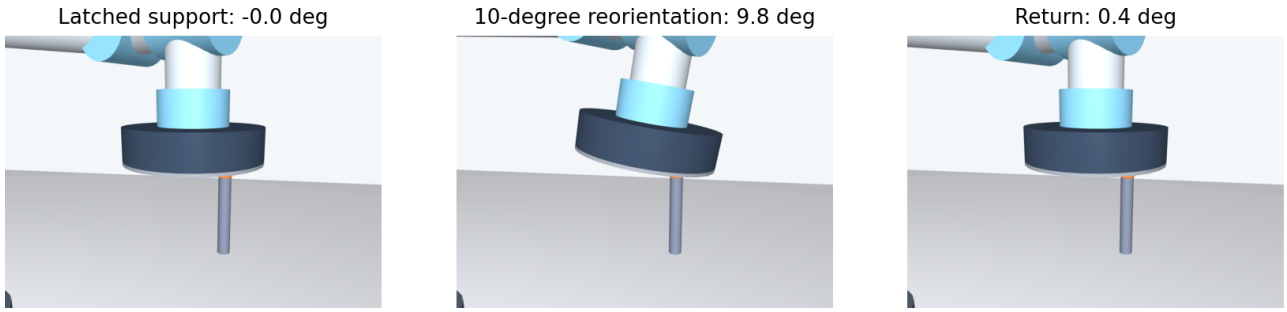


Figure 12. Rendered frames recovered from a saved +10-degree EIT trial at +30 mm. Labels give simulated achieved angles. The geometry is a simplified visualization of the same kinematic model, not a photograph or a separately choreographed animation.

Purpose and procedure of the numerical checks. Check whether the reported behaviour depends strongly on calculation resolution or errors in the implemented equations. Solve the electrical model with finer meshes, check its reciprocal drive/sense responses, compare the implemented contact-point Jacobian with small numerical position changes, and repeat three representative robot cases with the simulation time step halved. These checks test calculation consistency within the assumed models.

Numerical checks: electrical reciprocity relative error 6.5e-16; relative change in contact-induced voltages from 2.8 to 2.0 mm mesh spacing 0.52%; point-Jacobian finite-difference error 1.6e-07; maximum change from a 1 ms to 0.5 ms

robot step 0.0015 degrees and 0.0004 N across three representative controllers. No primary trial hits the force/speed guard or joint torque saturation. These checks establish numerical consistency, not physical fidelity.

10. Discussion: what can the paper claim?

Supported claim. Within this simulated contact family, using distributed electrical measurements to select where virtual mechanics act improves a concrete supported-reorientation task. The benefit persists with coarse location information and does not require EIT force feedback. A task-oriented evaluation is more informative than location RMSE alone, because retaining contact can coexist with failing the orientation goal.

Relationship to the draft. Spatial attachment, contact latching and bias sweeps already belong to original function D [1]. The new work adds physical arm dynamics, a synthetic electrical observation model, task success, timing costs and a sensing-quality study. It should not claim a new VMC principle, a new EIT reconstruction method, a human physiological result, or superiority over all other tactile sensors. No physiological experiment or direct human comparison is included.

Use case. The task represents a tool that uses an encountered support while changing orientation, for example aligning an inspection tool or adapting an approach pose against a fixture. The experiments establish this motion primitive, not successful assembly, cutting or insertion. Those applications require additional task geometry and failure criteria.

10.1 Limits of EIT realism

Eight electrodes do not identify the sensor physics. Prior hydrogel work can use a conductive overlay or grounded contact to alter current paths [5,6], while a piezoresistive or deforming sheet changes conductance differently. The present results use an ungrounded sheet with an assumed local conductance decrease. An optional distributed-shunt term exists in the solver but is not validated or used for the reported results. If TouchLoop relies on grounding, a multilayer contact resistance, capacitive coupling or geometry change, its calibration model must be changed before transfer claims are made.

The electrical sheet is not mechanically deformed by a continuum mechanics solution. Its Gaussian perturbation, width, saturation law, hysteresis and lag are synthetic. The front collision compliance does not represent measured elastomer stiffness, pad thickness, viscoelasticity or a distributed pressure map. The model supports a single compact contact; multiple contacts, edge contact, shear-dependent voltages and sensor bending are untested. Current excitation, ADC range, electrode impedance and acquisition protocol must be measured. The 37-45 mm radial band is outside the estimator training distribution.

10.2 Limits of control and evaluation

The arm has ideal torque actuation, exact bias compensation and simplified arm geometry. Arm self-collision and all real fixture clearances are not validated. There is no formal passivity proof. Primary rotations are small, trials vary synthetic sensing sessions more than robot mechanics, and most contact conditions stick. Success thresholds describe this proof of concept rather than an application specification. Five sensor seeds are not five independently built sensors; sample-wise confidence intervals would be misleading. Results are therefore reported as trial counts and effects, without treating trajectory samples as independent repetitions.

10.3 Hardware priorities

First measure the electrode drive/sense schedule, raw no-contact voltages, excitation frequency/amplitude and actual frame timestamps. Then gather held-out contact episodes with the deadweight attachment across location, load, loading direction and sessions. Identify the material law and noise before replacing the synthetic estimator. Validate the mechanics with a known-location reference, then compare fixed and EIT attachments on the same support. Establish force-feedback capability separately using an independent load reference. Recheck timing and approach forces before attempting the pivot on hardware.

11. Reproducibility and evidence status

The project contains all source code, the pinned official robot parameter files and their license, trained synthetic estimator, calibration datasets, frozen trial schedules, trial-level metrics, sampled trajectories, numerical checks, figures and this report. No real robot commands are present. The project is available at /media/yi/Yi Cam data/09_EIT_vmc/ur3e_sim. The S1-S4 scene figures are reconstructed from saved trial states.

Artifact	Purpose
src/eit.py	CEM forward solver, synthetic calibration, learned estimator
src/robot.py; src/experiments.py	Official-parameter UR3e model and S0-S4 controllers
src/run_suite.py; src/run_extension.py	Primary and exploratory protocols; resumable primary trials
src/audit_events.py	First-trigger audit and contact-event correction record
src/analyze.py; src/figures.py; src/make_report.py	Episode summaries, setup/results figures and report
results/trial_data/	619 robot episode traces sampled at 100 Hz
results/eit_*.npz	Separate calibration / validation / test sessions
results/audit/	Pre-correction source and affected-trial audit
report/figures/	PNG and vector PDF figures

A post-run implementation audit found that an initial electrical trigger was being ignored when physical contact was absent. This would hide false acquisition in the highest-noise tests. The gate was removed, every unique pre-latch history was audited, and affected trials were replaced by failed-acquisition results. The original source and the correction record are retained. The primary task settings were not retuned. Re-running the current code directly produces the corrected event semantics.

References

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[4] A Coupled Double-Layer Electrical Impedance Tomography-Based Sensing Skin for Pressure and Leak Detection. Sensors, 2024. https://pmc.ncbi.nlm.nih.gov/articles/PMC11244238/. Used as background for CEM-based sensing, not as a source of this sensor's material parameters.

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[6] X. Xu, D. Hardman and F. Iida. Economy of Touch: Task-Driven Information Selection in Electrical Impedance Tomography-based Tactile Robotic Sensing. Advanced Intelligent Systems, 2026. https://doi.org/10.1002/aisy.202501493. Eight-electrode hydrogel acquisition is a related example, not a validated specification of this device.

AI assistance: OpenAI Codex generated and executed the simulation code, checked numerical behaviour, and drafted this report. All plotted values come from the accompanying numerical outputs. No hardware measurements are presented. Human review of model assumptions and controller transfer is required before publication or robot use.